

API Reference

Cheap Public Transport API

API Version: 0.1.0

This API find the cheapest journey between UK cities via public transport. This combines national rail services, FlixBus coaches, and National Express coaches.

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Security and Authentication

SECURITY SCHEMES

KEY	TYPE	DESCRIPTION
HTTPBearer	http, bearer	

API

1. AUTHENTICATION

Endpoints to authenticate users via Supabase.

1.1 POST /auth/signup

Signup

Register a new account

Creates a new user record in Supabase with a default 'user' role.

Args:

- **data:** Schema containing the user's email and password.

Returns:

- A success message and the unique Supabase user ID.

Errors:

- **400:** If the email is already registered or the password does not meet requirements.

REQUEST

REQUEST BODY - application/json

```
{
  email*    string
  password* string
}
```

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```
{
  message* string
  id*      string
}
```

STATUS CODE - 422: Validation Error

RESPONSE MODEL - application/json

```
{
  detail [{
    Array of object:
    loc*
    ANY OF
    prop0
    string
    prop1
  }
}
```

```

        integer
    msg*   string
    type*  string
    input  undefined
    ctx {
    }
  }
}

```

1.2 POST /auth/login

Login

Authenticate user

Verifies credentials and returns a set of JWT tokens for subsequent requests.

Args:

- **data:** Schema containing the user's email and password.

Returns:

- **access_token:** The Bearer token used for authorization headers.
- **refresh_token:** Used to obtain a new access token when it expires.
- **token_type:** Always returns 'bearer'.

Errors:

- **401:** If the email or password credentials are incorrect.

REQUEST

REQUEST BODY - application/json

```

{
  email*   string
  password* string
}

```

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```

{
  access_token* string
  refresh_token* string
  token_type    string DEFAULT:bearer
}

```

STATUS CODE - 422: Validation Error

RESPONSE MODEL - application/json

```

{
  detail [{
    Array of object:
    loc*

```

```
    ANY OF
    prop0
    string
    prop1
    integer
    msg*  string
    type* string
    input undefined
    ctx {
    }
  }
}
```

2. CITIES

The 200 largest UK cities with latitude and longitude.

2.1 GET /city/

Get All Cities

Get all cities

Retrieves a list of every city currently stored in the database.

Access Level: - Admin and User.

Args:

- **db:** Database session dependency.

Returns:

- A list of City objects.

Errors:

- **401:** Unauthorized - Missing or invalid authentication token.
- **403:** Forbidden - User role not recognized.

REQUEST

No request parameters

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```
[ {  
  Array of object:  
    name*      string  
    latitude*   string  PATTERN:^(?!^[-+.]*)$[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.  
                        \d{0,6}0*$)  
    longitude*  string  PATTERN:^(?!^[-+.]*)$[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.  
                        \d{0,6}0*$)  
    id*         integer  
  } ]
```

2.2 POST /city/

Create New City

Create a new city

Adds a unique city record to the database.

Access Level: - Admin only.

Args:

- **city:** Schema containing name, latitude, and longitude.
- **db:** Database session dependency.

Errors:

- **401:** Unauthorized - Missing or invalid authentication token.
- **403:** Forbidden - User does not have administrative privileges.
- **422:** Validation Error - Input data is invalid.

Notes:

- City names are automatically converted to lowercase.
- The name field must be unique.

REQUEST

REQUEST BODY - application/json

```
{
  name*                string
  latitude*
    ANY OF
    prop0
    number
    prop1
    string
    PATTERN:^(?!^[-+.*$)[-+]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\d{0,6}0*$)
  longitude*
    ANY OF
    prop0
    number
    prop1
    string
    PATTERN:^(?!^[-+.*$)[-+]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\d{0,6}0*$)
}
```

RESPONSE

STATUS CODE - 201: Successful Response

RESPONSE MODEL - application/json

```
{
  name*      string
  latitude*   string  PATTERN:^(?!^[-+.*$)[-+]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\d{0,6}0*$)
  longitude* string  PATTERN:^(?!^[-+.*$)[-+]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\d{0,6}0*$)
  id*        integer
}
```

STATUS CODE - 422: Validation Error

RESPONSE MODEL - application/json

```
{
  detail [{
```


Array of object:

```
loc*
  ANY OF
  prop0
  string
  prop1
  integer
msg*  string
type* string
input undefined
ctx {
}
}]
}
```

2.3 GET /city/id/{city_id}

Get City By Id

Get city by ID

Fetches a specific city's details using its unique integer ID.

Access Level: - Admin and User.

Args:

- **city_id:** The unique identifier of the city.
- **db:** Database session dependency.

Errors:

- **401:** Unauthorized - Missing or invalid authentication token.
- **403:** Forbidden - User role not recognized.
- **404:** Not Found - No city exists with the provided ID.

REQUEST

PATH PARAMETERS

NAME	TYPE	DESCRIPTION
*city_id	integer	

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```
{
  name*      string
  latitude*  string  PATTERN:^(?!^[ -+.*$][+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.
\d{0,6}0*$)
  longitude* string  PATTERN:^(?!^[ -+.*$][+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.
\d{0,6}0*$)
  id*        integer
```

```
}
```

STATUS CODE - 404: City not found

STATUS CODE - 422: Validation Error

RESPONSE MODEL - application/json

```
{
  detail [{
    Array of object:
    loc*
      ANY OF
      prop0
      string
      prop1
      integer
    msg* string
    type* string
    input undefined
    ctx {
    }
  }]
}
```

2.4 GET /city/name/{city_name}

Get City By Name

Get city by name

Fetches a specific city's details using its name.

Access Level: - Admin and User.

Args:

- city_name: The name string of the city.
- db: Database session dependency.

Errors:

- 401: Unauthorized - Missing or invalid authentication token.
- 403: Forbidden - User role not recognized.
- 404: Not Found - No city exists with the provided name.

REQUEST

PATH PARAMETERS

NAME	TYPE	DESCRIPTION
*city_name	string	

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```
{
  name*      string
  latitude*   string  PATTERN:^(?!^[-+.]*$)[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.
                    \d{0,6}0*$)
  longitude*  string  PATTERN:^(?!^[-+.]*$)[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.
                    \d{0,6}0*$)
  id*         integer
}
```

STATUS CODE - 404: City not found

STATUS CODE - 422: Validation Error

RESPONSE MODEL - application/json

```
{
  detail [{
    Array of object:
    loc*
    ANY OF
    prop0
    string
    prop1
    integer
    msg*  string
    type* string
    input undefined
    ctx {
    }
  }]
}
```

2.5 PUT /city/{city_id}

Update City

Update city

Modifies an existing city record based on its ID.

Access Level: - Admin only.

Args:

- **city_id:** The unique identifier of the city to update.
- **city_update:** The new data to apply to the city.
- **db:** Database session dependency.

Errors:

- **401:** Unauthorized - Missing or invalid authentication token.
- **403:** Forbidden - User does not have administrative privileges.
- **404:** Not Found - The city to update is not found.

REQUEST

PATH PARAMETERS

NAME	TYPE	DESCRIPTION
*city_id	integer	

REQUEST BODY - application/json

```
{
  name*                               string
  latitude*
    ANY OF
    prop0
    number
    prop1
    string
    PATTERN:(?!^[ -+.*$)[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\d{0,6}0*$)
  longitude*
    ANY OF
    prop0
    number
    prop1
    string
    PATTERN:^(?!^[ -+.*$)[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\d{0,6}0*$)
}
```

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```
{
  name*      string
  latitude*   string  PATTERN:^(?!^[ -+.*$)[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\d{0,6}0*$)
  longitude*  string  PATTERN:^(?!^[ -+.*$)[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\d{0,6}0*$)
  id*        integer
}
```

STATUS CODE - 404: City not found

STATUS CODE - 422: Validation Error

RESPONSE MODEL - application/json

```
{
  detail [{
    Array of object:
    loc*
      ANY OF
      prop0
      string
      prop1
      integer
    msg*  string
    type* string
    input undefined
    ctx {
    }
```

}
}]

3. JOURNEYS

A collection of Travel Routes which makes a journey between any two given cities. This will return the cheapest possible combination of Travel Routes to complete a journey, as well as cost breakdown.

3.1 GET /journeys/id

Get Journey

Fetch a journey by city IDs

Calculates the cheapest route between two cities using their unique integer IDs.

Access Level: - Admin and User.

Args:

- **origin_id:** ID of the starting city.
- **destination_id:** ID of the destination city.
- **railcard_discount:** Percentage discount to apply (default 0).
- **advanced_fares:** Whether to include advanced fare pricing estimate.
- **max_coach_legs:** Maximum number of coach segments allowed.

Returns:

- A breakdown of total costs and a list of journey legs.

Errors:

- **401:** Unauthorized - Missing or invalid authentication token.
- **403:** Forbidden - User role not recognized.
- **404:** Not Found - No journey could be calculated between the provided IDs.

Notes:

- Total price includes a £2 changeover fee for station transfers.
- Returns the full path including all journey legs.

REQUEST

QUERY PARAMETERS

NAME	TYPE	DESCRIPTION
*origin_id	integer	
*destination_id	integer	
railcard_discount	integer	
advanced_fares	boolean	
max_coach_legs	integer	

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```
{
  total_price*      string      PATTERN:^(?!^[-+.]*)[+-]?0*\d*\.\?\d*$
  ticket_price*     string      PATTERN:^(?!^[-+.]*)[+-]?0*\d*\.\?\d*$
  railcard_discounts* string    PATTERN:^(?!^[-+.]*)[+-]?0*\d*\.\?\d*$
  advanced_discounts* string    PATTERN:^(?!^[-+.]*)[+-]?0*\d*\.\?\d*$
  transfer_price*   string      PATTERN:^(?!^[-+.]*)[+-]?0*\d*\.\?\d*$
  path* [{
    Array of object:
      price*        string      PATTERN:^(?!^[-+.]*)[+-]?0*(?:\d{0,8}|(?:=[\d.]{1,11}0*$)\d{0,8}\.\d{0,2}0*$)

      notes*
        ANY OF
        prop0
        string
        prop1
        null
      id*            integer
      origin_station* {
        name*        string
        city_id*      integer
        city* {
          name*       string
          latitude*   string    PATTERN:^(?!^[-+.]*)[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\d{0,6}0*$)
          longitude*  string    PATTERN:^(?!^[-+.]*)[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\d{0,6}0*$)
          id*         integer
        }
      }
      destination_station* {
        name*        string
        city_id*      integer
        city* {
          name*       string
          latitude*   string    PATTERN:^(?!^[-+.]*)[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\d{0,6}0*$)
          longitude*  string    PATTERN:^(?!^[-+.]*)[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\d{0,6}0*$)
          id*         integer
        }
      }
      transport_mode* {
        name* string
        id*  integer
      }
    }]
}
```

STATUS CODE - 422: Validation Error

RESPONSE MODEL - application/json

```
{
  detail [{
    Array of object:
      loc*
      ANY OF
```

```

    prop0
    string
    prop1
    integer
    msg*  string
    type* string
    input undefined
    ctx {
    }
  }
}

```

3.2 GET /journeys/name

Get Journey

Fetch a journey by city names

Calculates the cheapest route between two cities using their string names.

Access Level: - Admin and User.

Args:

- **origin_name:** Name of the starting city (case-insensitive).
- **destination_name:** Name of the destination city (case-insensitive).
- **railcard_discount:** Percentage discount to apply (default 0).
- **advanced_fares:** Whether to include advanced fare pricing estimate.
- **max_coach_legs:** Maximum number of coach segments allowed.

Returns:

- A breakdown of total costs and a list of journey legs.

Errors:

- **401:** Unauthorized - Missing or invalid authentication token.
- **403:** Forbidden - User role not recognized.
- **404:** Not Found - If either city name does not exist or no path is found.

Notes:

- Total price includes a £2 changeover fee for station transfers.

REQUEST

QUERY PARAMETERS

NAME	TYPE	DESCRIPTION
*origin_name	string	
*destination_name	string	
railcard_discount	integer	
advanced_fares	boolean	
max_coach_legs	integer	

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```
{
  total_price*      string      PATTERN:^(?!^[-+.]*)[+-]?0*\d*\.\?\d*$
  ticket_price*     string      PATTERN:^(?!^[-+.]*)[+-]?0*\d*\.\?\d*$
  railcard_discounts* string    PATTERN:^(?!^[-+.]*)[+-]?0*\d*\.\?\d*$
  advanced_discounts* string    PATTERN:^(?!^[-+.]*)[+-]?0*\d*\.\?\d*$
  transfer_price*   string      PATTERN:^(?!^[-+.]*)[+-]?0*\d*\.\?\d*$
  path* [{
    Array of object:
      price*        string      PATTERN:^(?!^[-+.]*)[+-]?0*(?:\d{0,8}|(?:=[\d.]{1,11}0*$)\d{0,8}\.\.\d{0,2}0*$)

      notes*
        ANY OF
        prop0
        string
        prop1
        null
      id*            integer
      origin_station* {
        name*        string
        city_id*      integer
        city* {
          name*       string
          latitude*   string    PATTERN:^(?!^[-+.]*)[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\.\d{0,6}0*$)
          longitude*  string    PATTERN:^(?!^[-+.]*)[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\.\d{0,6}0*$)
          id*         integer
        }
      }
      destination_station* {
        name*        string
        city_id*      integer
        city* {
          name*       string
          latitude*   string    PATTERN:^(?!^[-+.]*)[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\.\d{0,6}0*$)
          longitude*  string    PATTERN:^(?!^[-+.]*)[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\.\d{0,6}0*$)
          id*         integer
        }
      }
      transport_mode* {
        name* string
        id*  integer
      }
    }]
}
```

STATUS CODE - 422: Validation Error

RESPONSE MODEL - application/json

```
{
```

```
detail [{
```

Array of object:

```
  loc*
```

```
    ANY OF
```

```
    prop0
```

```
    string
```

```
    prop1
```

```
    integer
```

```
  msg*  string
```

```
  type*  string
```

```
  input  undefined
```

```
  ctx {
```

```
  }
```

```
}]
```

```
}
```

4. STATIONS

4.1 GET /stations/

Get All Stations

Get all stations

Retrieves a complete list of all stations stored in the database.

Access Level: - Admin and User.

Args:

- **db:** Database session dependency.

Returns:

- A list of Station objects.

Errors:

- **401:** Unauthorized - Missing or invalid authentication token.
- **403:** Forbidden - User role not recognized.

REQUEST

No request parameters

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```
[{  
  Array of object:  
  name*           string  
  city_id*        integer  
  city* {  
    name*         string  
    latitude*     string  PATTERN:^(?!^[-+.]*$)[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\d{0,6}0*$)  
    longitude*    string  PATTERN:^(?!^[-+.]*$)[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\d{0,6}0*$)  
    id*           integer  
  }  
}]
```

4.2 POST /stations/

Create New Station

Create a new station

Adds a new station record to the database after verifying the associated city exists.

Access Level: - Admin only.

Args:

- **station:** Schema containing the station name and city ID.
- **db:** Database session dependency.

Errors:

- **401:** Unauthorized - Missing or invalid authentication token.
- **403:** Forbidden - User does not have administrative privileges.
- **404:** Not Found - If the provided city_id does not match any existing city.

REQUEST

REQUEST BODY - application/json

```
{
  name*      string
  city_id*   integer
}
```

RESPONSE

STATUS CODE - 201: Successful Response

RESPONSE MODEL - application/json

```
{
  name*      string
  city_id*   integer
  city* {
    name*     string
    latitude* string PATTERN:^(?!^[-+.]*$)[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11})0*$)\d{0,4}\.\d{0,6}0*$)
    longitude* string PATTERN:^(?!^[-+.]*$)[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11})0*$)\d{0,4}\.\d{0,6}0*$)
    id*       integer
  }
}
```

STATUS CODE - 404: City not found

STATUS CODE - 422: Validation Error

RESPONSE MODEL - application/json

```
{
  detail [{
    Array of object:
    loc*
    ANY OF
    prop0
    string
    prop1
    integer
    msg* string
    type* string
    input undefined
  }
}
```

```

    ctx {
    }
  }
}

```

4.3 GET /stations/{station_id}

Get Station By Id

Get station by ID

Fetches the details of a specific station using its unique integer ID.

Access Level: - Admin and User.

Args:

- **station_id:** The unique identifier of the station.
- **db:** Database session dependency.

Errors:

- **401:** Unauthorized - Missing or invalid authentication token.
- **403:** Forbidden - User role not recognized.
- **404:** Not Found - If no station exists with the provided ID.

REQUEST

PATH PARAMETERS

NAME	TYPE	DESCRIPTION
*station_id	integer	

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```

{
  name*                string
  city_id*             integer
  city* {
    name*              string
    latitude*          string  PATTERN:^(?!^[ -+.*$][+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11})0*$)\d{0,4}\.\d{0,6}0*$)
    longitude*         string  PATTERN:^(?!^[ -+.*$][+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11})0*$)\d{0,4}\.\d{0,6}0*$)
    id*                integer
  }
}

```

STATUS CODE - 404: Station not found

STATUS CODE - 422: Validation Error

RESPONSE MODEL - application/json

```
{
  detail [{
    Array of object:
    loc*
    ANY OF
    prop0
    string
    prop1
    integer
    msg*  string
    type* string
    input undefined
    ctx {
    }
  }]
}
```

5. TRANSPORT MODES

Categorises routes based on transport mode such as 'train' or 'coach'.

5.1 GET /transport_mode/

Get All Transport Mode

Retrieve all transport modes

Fetches a complete list of all available transport categories.

Access Level: - Admin and User.

Args:

- **db:** Database session dependency.

Returns:

- A list of TransportMode objects, useful for UI filters or dropdown menus.

Errors:

- **401:** Unauthorized - Missing or invalid authentication token.
- **403:** Forbidden - User role not recognized.

REQUEST

No request parameters

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```
[ {  
  Array of object:  
    name*  string  
    id*    integer  
  } ]
```

5.2 POST /transport_mode/

Create New Transport Mode

Register a new transport mode

Adds a new category of transport (e.g., Train, Coach) to the system.

Access Level: - Admin only.

Args:

- **transport_mode**: Schema containing the name of the transport mode.
- **db**: Database session dependency.

Errors:

- **401**: Unauthorized - Missing or invalid authentication token.
- **403**: Forbidden - User does not have administrative privileges.

Notes:

- This creates a parent category that individual routes must be associated with.

REQUEST

REQUEST BODY - application/json

```
{
  name*  string
}
```

RESPONSE

STATUS CODE - 201: Successful Response

RESPONSE MODEL - application/json

```
{
  name*  string
  id*    integer
}
```

STATUS CODE - 422: Validation Error

RESPONSE MODEL - application/json

```
{
  detail [{
    Array of object:
    loc*
    ANY OF
    prop0
    string
    prop1
    integer
    msg*  string
    type* string
    input undefined
    ctx {
    }
  }]
}
```

5.3 GET /transport_mode/{transport_mode_id}

Get Transport Mode By Id

Get transport mode by ID

Retrieves the details of a specific transport category using its unique ID.

Access Level: - Admin and User.

Args:

- **transport_mode_id:** The unique identifier of the transport mode.
- **db:** Database session dependency.

Errors:

- **401:** Unauthorized - Missing or invalid authentication token.
- **403:** Forbidden - User role not recognized.
- **404:** Not Found - If no transport mode exists with the provided ID.

REQUEST

PATH PARAMETERS

NAME	TYPE	DESCRIPTION
*transport_mode_id	integer	

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```
{
  name* string
  id* integer
}
```

STATUS CODE - 422: Validation Error

RESPONSE MODEL - application/json

```
{
  detail [{
    Array of object:
    loc*
    ANY OF
    prop0
    string
    prop1
    integer
    msg* string
    type* string
    input undefined
    ctx {
    }
  }]
}
```

6. TRAVEL ROUTES

The individual intercity legs which make up a Journey.

6.1 GET /routes/

Get All Routes

List all travel routes

Retrieves every available route between stations from the database.

Access Level: - Admin and User.

Args:

- **db:** Database session dependency.

Returns:

- A list of all Route objects.

Errors:

- **401:** Unauthorized - Missing or invalid authentication token.
- **403:** Forbidden - User role not recognized.

Notes:

- **Caution:** This dataset contains ~20,000 records. Testing this in a browser may cause performance issues or crashes.

REQUEST

No request parameters

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```
[{  
  Array of object:  
    price*      string      PATTERN:^(?!^[ -+.*$])[+-]?0*(?:\d{0,8}|(?:=[\d.]{1,11}0*$  
    \d{0,8})\.\d{0,2}0*$)  
    notes*  
      ANY OF  
      prop0  
      string  
      prop1  
      null  
    id*          integer  
    origin_station* {  
      name*      string  
      city_id*   integer
```

```

    city* {
      name*      string
      latitude*   string  PATTERN:^(?!^[-+.*$)[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\d{0,6}0*$)
      longitude*  string  PATTERN:^(?!^[-+.*$)[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\d{0,6}0*$)
      id*         integer
    }
  }
  destination_station* {
    name*      string
    city_id*    integer
    city* {
      name*      string
      latitude*   string  PATTERN:^(?!^[-+.*$)[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\d{0,6}0*$)
      longitude*  string  PATTERN:^(?!^[-+.*$)[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\d{0,6}0*$)
      id*         integer
    }
  }
  transport_mode* {
    name*      string
    id*        integer
  }
}]]

```

6.2 POST /routes/

Create New Route

Create a new travel route

Adds a new route record after validating that the stations and transport mode exist.

Access Level: - Admin only.

Args:

- **route:** Schema containing origin/destination station IDs, transport mode ID, and price.
- **db:** Database session dependency.

Errors:

- **401:** Unauthorized - Missing or invalid authentication token.
- **403:** Forbidden - User does not have administrative privileges.
- **404:** Not Found - If the origin station, destination station, or transport mode does not exist.

REQUEST

REQUEST BODY - application/json

```

{
  price*
  ANY OF
  prop0
  number

```

```

prop1
string
PATTERN:^(?!^[-+.*$][+-]?0*(?:\d{0,8}|(?:=[\d.]{1,11}0*$)\d{0,8}\.\d{0,2}0*$)
notes*
  ANY OF
  prop0
  string
  prop1
  null
origin_station_id*      integer
destination_station_id* integer
transport_mode_id*      integer
}

```

RESPONSE

STATUS CODE - 201: Successful Response

RESPONSE MODEL - application/json

```

{
  price*      string      PATTERN:^(?!^[-+.*$][+-]?0*(?:\d{0,8}|(?:=[\d.]{1,11}0*
    $)\d{0,8}\.\d{0,2}0*$)

  notes*
    ANY OF
    prop0
    string
    prop1
    null
  id*          integer
  origin_station* {
    name*      string
    city_id*   integer
    city* {
      name*    string
      latitude* string PATTERN:^(?!^[-+.*$][+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*
        $)\d{0,4}\.\d{0,6}0*$)
      longitude* string PATTERN:^(?!^[-+.*$][+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*
        $)\d{0,4}\.\d{0,6}0*$)
      id*      integer
    }
  }
  destination_station* {
    name*      string
    city_id*   integer
    city* {
      name*    string
      latitude* string PATTERN:^(?!^[-+.*$][+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*
        $)\d{0,4}\.\d{0,6}0*$)
      longitude* string PATTERN:^(?!^[-+.*$][+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*
        $)\d{0,4}\.\d{0,6}0*$)
      id*      integer
    }
  }
  transport_mode* {
    name* string
    id*   integer
  }
}

```

```
}
```

STATUS CODE - 422: Validation Error

RESPONSE MODEL - application/json

```
{
  detail [{
    Array of object:
    loc*
    ANY OF
    prop0
    string
    prop1
    integer
    msg* string
    type* string
    input undefined
    ctx {
    }
  }]
}
```

6.3 GET /routes/{route_id}

Get Route By Id

Get specific route details

Fetches the full details of a single travel route using its unique ID.

Access Level: - Admin and User.

Args:

- route_id: The unique identifier of the route.
- db: Database session dependency.

Errors:

- 401: Unauthorized - Missing or invalid authentication token.
- 403: Forbidden - User role not recognized.
- 404: Not Found - No route exists with the provided ID.

REQUEST

PATH PARAMETERS

NAME	TYPE	DESCRIPTION
*route_id	integer	

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```

{
  price*      string      PATTERN:^(?!^[-+.*$][+-]?0*(?:\d{0,8}|(?:=[\d.]{1,11}0*$)\d{0,8}\.\d{0,2}0*$)

  notes*
    ANY OF
    prop0
    string
    prop1
    null
  id*         integer
  origin_station* {
    name*      string
    city_id*   integer
    city* {
      name*     string
      latitude* string PATTERN:^(?!^[-+.*$][+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\d{0,6}0*$)
      longitude* string PATTERN:^(?!^[-+.*$][+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\d{0,6}0*$)
      id*       integer
    }
  }
  destination_station* {
    name*      string
    city_id*   integer
    city* {
      name*     string
      latitude* string PATTERN:^(?!^[-+.*$][+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\d{0,6}0*$)
      longitude* string PATTERN:^(?!^[-+.*$][+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\d{0,6}0*$)
      id*       integer
    }
  }
  transport_mode* {
    name*      string
    id*        integer
  }
}

```

STATUS CODE - 404: Route not found

STATUS CODE - 422: Validation Error

RESPONSE MODEL - application/json

```

{
  detail [{
    Array of object:
    loc*
    ANY OF
    prop0
    string
    prop1
    integer
  }
  msg*  string
  type* string
  input undefined
  ctx {

```

```
}
  }
}
```

6.4 PUT /routes/{route_id}

Update Route

Update an entire route

Replaces an existing route's data. Validates the existence of related entities before saving.

Access Level: - Admin only.

Args:

- **route_id**: The unique identifier of the route to update.
- **route_update**: The new data to apply to the route.
- **db**: Database session dependency.

Errors:

- **401**: Unauthorized - Missing or invalid authentication token.
- **403**: Forbidden - User does not have administrative privileges.
- **404**: Not Found - If the route, either station, or the transport mode is not found.

REQUEST

PATH PARAMETERS

NAME	TYPE	DESCRIPTION
*route_id	integer	

REQUEST BODY - application/json

```
{
  price*
  ANY OF
  prop0
  number
  prop1
  string
  PATTERN:^(?!^[-+.]*$)[+-]?0*(?:\d{0,8}|(?:=[\d.]{1,11}0*$)\d{0,8}\.\d{0,2}0*$)
  notes*
  ANY OF
  prop0
  string
  prop1
  null
  origin_station_id*           integer
  destination_station_id*      integer
  transport_mode_id*           integer
}
```

RESPONSE

STATUS CODE - 200: Successful Response

RESPONSE MODEL - application/json

```
{
  price*      string      PATTERN:^(?!^[ -+.*$])[+-]?0*(?:\d{0,8}|(?:=[\d.]{1,11}0*$)\d{0,8}\.\d{0,2}0*$)

  notes*
    ANY OF
    prop0
    string
    prop1
    null
  id*         integer
  origin_station* {
    name*      string
    city_id*   integer
    city* {
      name*    string
      latitude* string PATTERN:^(?!^[ -+.*$])[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\d{0,6}0*$)
      longitude* string PATTERN:^(?!^[ -+.*$])[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\d{0,6}0*$)
      id*      integer
    }
  }
  destination_station* {
    name*      string
    city_id*   integer
    city* {
      name*    string
      latitude* string PATTERN:^(?!^[ -+.*$])[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\d{0,6}0*$)
      longitude* string PATTERN:^(?!^[ -+.*$])[+-]?0*(?:\d{0,4}|(?:=[\d.]{1,11}0*$)\d{0,4}\.\d{0,6}0*$)
      id*      integer
    }
  }
  transport_mode* {
    name*      string
    id*        integer
  }
}
```

STATUS CODE - 404: Route not found

STATUS CODE - 422: Validation Error

RESPONSE MODEL - application/json

```
{
  detail [{
    Array of object:
    loc*
    ANY OF
    prop0
    string
    prop1
    integer
    msg*  string
  }
}
```



```

    type* string
    input undefined
    ctx {
    }
  }
}

```

6.5 DELETE /routes/{route_id}

Delete Route

Delete a route

Permanently removes a specific route record from the database.

Access Level: - Admin only.

Args:

- **route_id:** The unique identifier of the route to be deleted.
- **db:** Database session dependency.

Errors:

- **401:** Unauthorized - Missing or invalid authentication token.
- **403:** Forbidden - User does not have administrative privileges.
- **404:** Not Found - No route exists with the provided ID.

Notes:

- This action is destructive but does not affect associated station or city records.

REQUEST

PATH PARAMETERS

NAME	TYPE	DESCRIPTION
*route_id	integer	

RESPONSE

STATUS CODE - 204: Successful Response

STATUS CODE - 404: Route not found

STATUS CODE - 422: Validation Error

RESPONSE MODEL - application/json

```

{
  detail [{
    Array of object:
    loc*
    ANY OF
    prop0
    string
    prop1
  }
}

```

```
    integer
  msg*  string
  type* string
  input undefined
  ctx {
  }
}]
}
```
